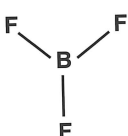
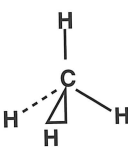
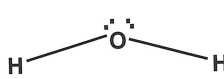
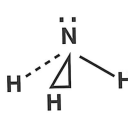


Set 18: Molecular shape

Notes

Polarity of a molecule is determined by the electron distribution in the bonds of the molecule and by the shape of a molecule. Shape is determined by the Valence Shell Electron Pair Repulsion Theory (VSEPR). This basically states that electrons repel each other and will position themselves as far from each other as possible. The shapes you should be familiar with are shown in the diagram below.

Triangular Planar		Bond angles for BF_3 are 120°
Tetrahedral		Bond angles for CH_4 are 109.5°
V-shaped		Bond angles for H_2O are 104.5°
Linear		Bond angles for BeCl_2 are 180°
Pyramidal		Bond angles for NH_3 are 107°

There are two requirements for a molecule to be polar:

- The molecule should have one or more polar bonds and
- The molecule should not be symmetrical.

Bonds are polar if the elements bonded have differing electronegativities (electron attracting power).

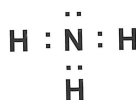
In the molecule of ammonia (NH_3) in the diagram above for example, nitrogen and hydrogen have differing electronegativities so polar bonds are present and the molecule shape is asymmetrical. This makes the ammonia molecule polar. Methane (CH_4) on the other hand has four polar bonds but the shape is symmetrical so the resulting molecule is non-polar.

Examples

When determining the shape and polarity of a molecule start by drawing an electron dot diagram.

1. Ammonia (NH_3): Nitrogen has 5 valence electrons. Hydrogen has one valence electron.

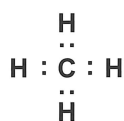
The electron dot diagram is:



Shape: As well as 3 bonding pairs of electrons there is a lone pair of electrons in the molecule and these also contribute to the shape of ammonia. The lone pair requires its share of space so the shape is **pyramidal** rather than triangular planar.

Polarity: In the molecule of ammonia (NH_3), nitrogen and hydrogen have differing electronegativities so polar bonds are present. The molecule shape is pyramidal (asymmetrical). This makes the ammonia molecule **polar**.

2. **Methane (CH_4):** Carbon has four valence electrons. Hydrogen has one valence electron. The carbon and hydrogen share electrons as shown in the electron dot diagram:



The molecule is symmetrical as the atoms space out evenly creating a **tetrahedral** shape. It has four polar bonds but the shape is symmetrical so the resulting methane molecule is **non-polar**.

Set 18: Exercises

- Explain why the electronegativity of iodine is much less than for fluorine.
- $$\begin{array}{l} \text{Cl} \text{-----} \text{Cl} \\ \text{Cl} \text{-----} \text{H} \\ \text{Cl} \text{-----} \text{Na} \end{array}$$
 - On the diagram above show by drawing a cross, where the shared pair of electrons would be most likely to be found between the two elements that are bonded. Explain why they would be in that position.
 - What is the significance of the position of the electrons to the bond formed?
- Which of the following species would have polar bonds?
 - I_2
 - CO_2
 - PH_3
 - CH_3Cl
 - O_2
- For the species in question 3, draw electron dot diagrams and determine their shape and their polarity.
- Draw diagrams to represent each of the five shapes of molecules discussed in this set. Which shapes are always going to be polar? Explain why.
- Use examples to explain how it is possible to have a non-polar molecule with polar bonds.
- How does the polarity affect the type of intermolecular force present?
- For the following species, draw the electron dot diagram and determine its shape.
 - SO_2
 - NH_3
 - H_2O
 - SO_3
 - NO_3^-
 - SO_4^{2-}
 - CO_3^{2-}
 - H_2O_2
 - C_2H_2
 - PCl_3